Notes: Chen, Noronha, and Singal (JF, 2004)

The Price Response to S&P 500 Index Additions and Deletions: Evidence of Asymmetry and a New Explanation

Contents

1 Big picture

- **Question.** What happens to prices when firms are added to or deleted from the S&P 500, and what mechanism explains the price response?
- **Main empirical puzzle.** Additions have a permanent positive price effect, but deletions do not have a permanent negative price effect.
- **Why this is puzzling.** Standard index-effect explanations usually predict symmetry: if index demand raises added-stock prices, index selling should lower deleted-stock prices.
- **Main result.** After index-fund trading becomes relevant, added firms earn large positive abnormal returns that persist after the effective date. Deleted firms suffer negative returns around announcement/effective dates, but these losses are largely reversed.
- **Rejected or incomplete explanations.** Pure price pressure, downward-sloping demand/imperfect substitutes, information, liquidity, and certification stories cannot easily explain why additions and deletions are asymmetric.
- **New explanation.** The paper proposes an investor awareness or investor recognition channel, drawing on Merton's incomplete-information idea.
- **Mechanism.** Index addition increases visibility, shareholder base, and institutional ownership. More investor recognition lowers the required return or "shadow cost" of holding the stock, raising price permanently.
- **Why deletion is different.** Deletion reduces index-fund demand but does not fully erase awareness. Deleted firms may remain known to investors, analysts, institutions, and past holders, so the permanent decline in recognition is weak.
- **Core evidence.** Shareholder counts and institutional ownership rise after additions. Merton-style shadow costs fall after additions. Changes in shadow cost are negatively related to long-window abnormal returns.
- **Takeaway.** S&P 500 index changes do not simply move prices through index-fund demand. They also change the investor base, and that change is asymmetric between additions and deletions.

2 Data and measurement

2.1 Event sample

- The sample contains all additions to and deletions from the S&P 500 between July 1962 and December 2000.
- Firms need return data for at least 60 trading days before the announcement date and 90 trading days after the effective date.
- The paper divides the sample into three periods:
  - **1962–August 1976:** before public pre-announcement of index changes and before meaningful index-fund demand.
  - **September 1976–September 1989:** pre-announcement period begins; index funds exist but are smaller.
  - **October 1989–December 2000:** S&P uses advance announcements and index assets are much larger.

Interpretation: The period split is central. The paper asks whether the price effect appears only once index changes become visible and investable events.

2.2 Event dates

- **Announcement date (AD):** first trading day following the announcement.
- **Effective date (ED):** date when the firm is actually added to or removed from the index.
- The paper studies returns on AD, from AD to ED, and from AD through post-effective windows.

Interpretation: The AD–ED window is especially important after 1989 because traders know the change before it becomes effective.

2.3 Abnormal returns

Abnormal returns are measured relative to the S&P 500 total return:

$$AR_{i,t} = R_{i,t} - R_{S\&P500,t}. \tag{1}$$

Cumulative abnormal returns are:

$$CAR_i(a,b) = \sum_{t=a}^{b} AR_{i,t}. \tag{2}$$

The paper reports:

- AD return;
- AD to ED return;
- $CAR20$: AD through ED plus 20 trading days;
- $CAR60$: AD through ED plus 60 trading days.

Interpretation: If price pressure is temporary, the initial abnormal return should reverse after ED. If the effect is permanent, $CAR20$ and $CAR60$ remain positive for additions or negative for deletions.

2.4 Turnover ratios

Let normal turnover be the average turnover in the 60 trading days before announcement. The paper reports event turnover ratios:

$$TurnoverRatio = \frac{\text{event-date turnover}}{\text{pre-event normal turnover}}. \tag{3}$$

It reports ratios for AD, ED, and the post-event window.

Interpretation: Turnover helps distinguish price pressure from permanent valuation effects. A large temporary turnover spike can occur even if prices eventually reverse.

2.5 Investor awareness measures

The paper uses two direct proxies for investor recognition:

- **Number of shareholders:** measured before announcement and at least nine months after effective date.
- **Institutional ownership:** number of institutional holders and percentage of shares held by institutions using 13F data.

Interpretation: If index addition increases investor recognition, it should expand the investor base. The paper therefore predicts more shareholders and more institutional ownership after additions.

2.6 Merton shadow cost

The paper computes a Merton-style shadow cost:

$$ShadowCost = \frac{ResidualStandardDev}{S\&P500MarketCap} \times \frac{FirmSize}{Number\ of\ Shareholders}. \tag{4}$$

- $ResidualStandardDev$ is the standard deviation of the difference between firm returns and S&P 500 returns.
- $FirmSize$ is the market value of equity.
- $Number\ of\ Shareholders$ proxies for the breadth of investor recognition.

Interpretation: A firm with fewer shareholders and higher idiosyncratic volatility has a higher shadow cost. More recognition lowers this cost and raises price.

3 Models and empirical specifications

3.1 Three benchmark hypotheses

1. **Price pressure:** index funds create temporary demand/supply pressure. Prices should reverse after the effective date.
2. **Imperfect substitutes/downward-sloping demand:** index demand permanently shifts prices. Additions and deletions should be roughly symmetric.
3. **Information/liquidity/certification:** index inclusion may reveal favorable information or improve liquidity. The challenge is explaining why deletion does not produce the corresponding negative effect.

Interpretation: The paper's asymmetry result is the key falsifying fact for symmetric demand-shock stories.

3.2 Investor recognition hypothesis

The investor recognition hypothesis says:

$$Index\ Addition \Rightarrow More\ investor\ awareness \Rightarrow Lower\ shadow\ cost \Rightarrow Higher\ price. \tag{5}$$

For deletions:

$$Index\ Deletion \not\Rightarrow Full\ loss\ of\ recognition. \tag{6}$$

Interpretation: Once a firm becomes known, deletion from the index does not necessarily make it unknown. This creates asymmetric permanent price effects.

3.3 Regression linking price effects to shadow cost

The paper estimates:

$$AR_i = \alpha + \beta(dShadow_i) + \epsilon_i, \tag{7}$$

where AR_i is the long-window abnormal return and $dShadow_i$ is the change in shadow cost.

It also estimates:

$$AR_i = \alpha + \beta_1 \log(RelSize_i) + \beta_2 \log(Age_i) + \beta_3 NYSEdum_i + \beta_4(dShadow_i) + \epsilon_i. \tag{8}$$

Interpretation: If investor awareness explains returns, β_4 should be negative. A lower shadow cost should be associated with a positive abnormal return.

4 Identification and empirical design

4.1 What identifies the main result

- The design compares event-window abnormal returns for additions and deletions.
- It compares early and later periods, exploiting changes in index-fund trading and S&P announcement practices.
- It studies whether price effects persist after the effective date, not just around AD or ED.
- It links price effects to changes in investor base and shadow cost.

Interpretation: The paper is not a randomized experiment. Its credibility comes from the conjunction of asymmetric return patterns, investor-base changes, and shadow-cost regressions.

4.2 Why deletion evidence is crucial

- Many earlier index studies focused on additions.
- Additions alone cannot distinguish demand pressure from investor awareness or information.
- Deletions provide the mirror event: if index demand is the key mechanism, deletions should produce permanent negative effects.
- The paper finds no such permanent negative effect.

Interpretation: Deletions are the paper's main identification lever. They reveal the asymmetry that motivates investor recognition.

4.3 Why shareholder and institutional ownership tests matter

- Investor recognition is otherwise difficult to observe directly.
- The number of shareholders gives a broad retail/investor-base measure.
- Institutional ownership gives an informed-investor and delegated-investor measure.
- Both are measured before and after index changes.

Interpretation: If prices rise because more investors become aware of the stock, the investor base should actually broaden. The paper documents exactly that for additions.

4.4 Limits of the design

- S&P index changes may contain some information.
- Firms are not randomly added or deleted.
- Shareholder data are noisy and not always available.
- Institutional holdings data begin only in 1980.
- The shadow-cost measure is constructed from several noisy components.

Interpretation: The paper's main claim is not that only investor awareness matters. It argues that investor awareness is necessary to explain the asymmetric pattern.

5 Tables and figures

The paper has no plotted figures. The central evidence is in six tables. The tables below are directly cropped from the paper and grouped compactly.

5.1 Table I: Abnormal returns and volume turnover around index changes

Read:

- Panel A shows that additions have little effect before 1976, but strong positive abnormal returns after public pre-announcement begins.
- During 1976–1989, additions earn a 3.171% announcement-date abnormal return and a 3.556% $CAR60$.
- During 1989–2000, additions earn a 5.446% announcement-date return, an 8.899% AD-to-ED return, and a 6.189% $CAR60$.
- Panel B shows that deletions have large negative announcement/effective-date returns in 1989–2000, including −8.462% on AD and −14.436% from AD to ED.
- But deletion $CAR60$ is only 0.394% in 1989–2000, implying strong reversal after the effective date.

Economic message: Additions have a permanent positive price effect; deletions have a temporary negative effect. This asymmetry is the central puzzle.

Panel A. Additions

	196207–197608	197609–198909	198910–200012
Initial sample	305	297	303
Final sample	279	263	218
Cumulative Abnormal Returns			
Anndate	−0.047	3.171***	5.446***
	0.495	0.932***	0.940***
Anndate to effdate			8.899***
			0.927***
Anndate to effdate + 20 (CAR20)	−0.742	3.123***	6.396***
	0.470	0.681***	0.688***
Anndate to effdate + 60 (CAR60)	0.588	3.556***	6.189***
	0.505	0.635***	0.615***
Turnover Ratios			
Anndate/Pre	0.763***	3.741***	3.703***
	0.363***	0.918***	0.991***
Effdate/Pre			12.323***
			0.995***
Post/Pre	0.906**	0.992	1.080***
	0.431**	0.492	0.601***

Panel B. Deletions

	196207–197608	197609–198909	198910–200012
Initial sample	305	297	303
Final sample	145	28	62
Cumulative Abnormal Returns			
Anndate	−0.407*	−1.168	−8.462***
	0.469	0.393	0.016***
Anndate to effdate			−14.436***
			0.032***
Anndate to effdate + 20 (CAR20)	1.189*	−1.642	−4.710
	0.593**	0.357	0.339**
Anndate to effdate + 60 (CAR60)	2.172	−1.715	0.394
	0.572*	0.429	0.452
Turnover Ratio			
Anndate/Pre	0.691***	3.523***	3.487***
	0.386***	0.857***	0.952***
Effdate/Pre			16.495***
			1.000***
Post/Pre	1.030	0.966	0.938
	0.531	0.429	0.403

*, **, and *** denote significance at the 10%, 5%, or 1% level, respectively.

5.2 Table II: Reconciling deletion results with prior papers

Read:

- Panel A reproduces Lynch and Mendenhall-style deletion returns and shows negative returns through ED plus 10 or 20 days.
- Panel B shows that the same deletion stocks rebound after the effective date.
- For the authors' sample, AD to ED is −12.5%, but ED to ED+40 is +25.8% and ED to ED+60 is +23.0%.
- Removing outliers weakens but does not eliminate the rebound.

Economic message: Earlier deletion studies correctly found short-run negative returns, but the longer-window evidence shows that deletions do not suffer a permanent price decline.

Reconciliation of Results for Deletions in Previous Papers

Panel A: With Lynch and Mendenhall (1997)

	Sample of Lynch and Mendenhall	Sample in This Paper	Explanation
1. Tested sample	15	16	Firms deleted between March 1990 and April 1995.
2. Announcement day (AD) abnormal return.	−6.26%	−7.82%***	All L-M returns are adjusted based on the CRSP value-weighted index until 1993, and on the S&P 500 daily return for 1994 and 1995. All our returns are adjusted for the S&P 500 daily return. The CARs with CRSP value-weighted index are smaller in magnitude.
3. CAR from AD to effective day (ED) + 10 trading days.	−14.44%	−18.30%***	
4. CAR from AD to ED + 20 trading days.	Not reported	−16.03%**	
5. CAR from AD to ED + 60 trading days.	Not reported	−6.32%	

Panel B: With Beneish and Whaley (2002)

CAR Period	Beneish and Whaley (2002) $N = 49$	Our Sample $N = 46$	Our Sample without Top Five Outliers, $N = 41$	Our Sample without Top Five and Bottom Five Outliers, $N = 36$
AD	#−9.9***	−8.9***	−7.7***	−7.6***
AD to ED	#−10.8***	−12.5***	−10.4***	−9.9***
ED to ED+20	+14.5**	+15.7**	+4.1	+7.2*
ED to ED+40	+23.7***	+25.8***	+6.9*	+12.8***
ED to ED+60	Not reported	+23.0**	+8.4	+15.7***
AD to ED+60	Not reported	+4.7	−3.1	+3.6

#These returns are calculated by the authors from the Beneish and Whaley paper, and not directly reported by them.
*, **, and *** denote significance at the 10%, 5%, or 1% level, respectively.

5.3 Table III: Number of shareholders

Read:

- Additions increase shareholder counts after index entry.
- In 1976–1989, added firms gain an average of 831 shareholders and a median of 145 shareholders.
- In 1989–2000, added firms gain an average of 2,936 shareholders and a median of 426 shareholders.
- Deletions show small or negative changes in shareholders, especially in 1989–2000.

Economic message: The investor base broadens after additions, consistent with investor awareness. Deletions do not show a similarly large permanent collapse in investor recognition.

Number of Shareholders

This table reports changes in the number of shareholders around S&P 500 index changes. The number of shareholders prior to the announcement date is taken at a time as late as possible but before the announcement date. The number of shareholders after the change is taken to be the number of shareholders at a time at least nine months after the effective date. Shareholder data are obtained from COMPUSTAT, Moody's Manuals, the S&P Market Encyclopedia, and from 10-K filings on SECs EDGAR, in that order.

Period	Sample Size in Table I	Sample Size for This Table	Mean (Median) of Pre-Event Number of Shareholders	Mean (Median) of Post-Event Number of Shareholders	Percent Change in Mean (Median)	Mean (Median) of Paired Changes
Panel A. Changes in Number of Shareholders—Additions						
197609—198909	263	243	17,728	18,560	4.7	831**
			8,568	10,000	16.7	145**
198910—200012	218	187	17,745	20,681	16.5	2,936**
			4,840	6,077	25.6	426***
Panel B. Changes in Number of Shareholders—Deletions						
197609—198909	28	20	25,584	25,192	−1.5	−391
			12,428	12,469	0.33	−515
198910—200012	62	54	7,870	7,464	−5.2	−406***
			6,046	5,504	−9.0	−215***

5.4 Table IV: Institutional ownership

Read:

- Additions increase both the number of institutional holders and the percentage of shares held by institutions.
- In 1980–1989, added firms have a mean paired increase of 19 institutional holders.
- In 1989–2000, added firms have a mean paired increase of 46.1 institutional holders.
- Deletions, especially in 1989–2000, lose institutional holders and institutional ownership share.

Economic message: Index additions expand institutional recognition and ownership. This is a direct empirical channel through which required returns can fall.

Table IV
Institutional Ownership

This table reports changes in the number of institutions and in the percentage of institutional shareholding around S&P 500 index changes. The number and fraction of shares held by institutions in the quarter immediately prior to the announcement date is compared with institutional holdings at least one quarter after the effective date. Institutional holdings data are obtained from 13f filings available from Thomson Financial from March 1980. Observations before March 1980 are excluded.

Period	Sample Size in Table I	Sample Size for This Table	Mean (Median) of Pre-Event Number of Institutions	Mean (Median) of Post-Event Number of Institutions	Percent Change in Mean (Median)	Mean (Median) of Paired Changes	Mean (Median) of Pre-Event Percent of Institutions	Mean (Median) of Post-Event Percent of Institutions	Mean (Median) of Paired Changes
Panel A. Change in Institutional Ownership—Additions									
198003—198909	221	203	96.2	115.2	19.7	19.0***	44.1	45.8	1.7***
			93.0	108.0	16.1	16.0***	43.2	45.6	1.6***
198910—200012	218	210	205.3	251.5	22.5	46.1***	60.0	61.1	1.1*
			204.0	237.5	16.4	38.0***	60.8	61.4	0.8***
Panel B. Change in Institutional Ownership—Deletions									
198003—198909	17	13	72.2	66.3	−8.2	−5.9*	25.8	24.7	−1.2
			56.5	40.5	−28.3	−3.5*	33.3	21.6	1.5
198910–200012	62	59	106.4	70.1	−34.1	−36.2***	52.7	46.0	−6.8***
			105.0	64.0	−39.1	−31.0***	54.6	48.1	−3.4***

*, **, and *** denote significance at the 10%, 5%, or 1% level, respectively.

5.5 Table V: Estimates of Merton's shadow cost

Read:

- For additions, the shadow cost falls significantly after index inclusion.
- In 1976–1989, mean shadow cost falls by 0.480×10^{-9} and median shadow cost falls by 0.095×10^{-9}.
- In 1989–2000, mean shadow cost falls by 3.581×10^{-9} and median shadow cost falls by 0.054×10^{-9}.
- For deletions, the shadow cost changes are small and often positive rather than negative.

Economic message: The shadow-cost evidence translates the awareness story into a pricing channel: additions reduce the cost of incomplete investor recognition.

Table V

Estimates of Merton's Shadow Cost

Merton's shadow cost is defined as below

$$ShadowCost = \frac{\text{ResidualStandardDev}}{\text{S \& P500MarketCap}} \times \frac{\text{Firm Size}}{\text{Number of Shareholders}},$$

where Firm Size (the market value of the firm's equity) and S&P 500 Market Cap are measured on the announcement day. The residual standard deviation is measured as the standard deviation of the difference between the firm's return and the total return of the S&P 500 index in the 252-day period before the index change announcement day (pre), and in the 252-day period after the effective day (post). The number of shareholders before the index change is obtained as late as possible prior to the announcement date. The number of shareholders after the change is obtained at least nine months after the effective date.

Period	Addition/ Deletion	Sample Size in Table I	Sample Size for This Table	Mean (Median) Shadow Cost before Change ($\times 10^9$)	Mean (Median) Shadow Cost after Change ($\times 10^9$)	Mean (Median) of Paired Changes
197609–198909	Additions	263	243	2.175	1.695	−0.480***
				1.491	1.337	−0.095***
	Deletions	28	20	0.283	0.313	0.030
				0.227	0.247	0.010
198910–200012	Additions	218	187	11.537	7.956	−3.581***
				3.595	3.040	−0.054***
	Deletions	62	54	0.450	0.563	0.112*
				0.256	0.278	0.055***

*, **, and *** denote significance at the 10%, 5%, or 1% level, respectively.

before the index change announcement for the pre period, and in the 252 day

5.6 Table VI: Abnormal returns and changes in shadow cost

Read:

- The dependent variable is long-window abnormal return, $CAR60$.
- The key coefficient is on $dShadow$, the change in shadow cost.
- For additions, the $dShadow$ coefficient is negative and statistically significant in both post-1976 periods.
- In 1976–1989, the addition coefficient is about -1.899 without controls and -1.846 with controls.
- In 1989–2000, the addition coefficient is about -0.501 without controls and -0.447 with controls.
- For 1989–2000 deletions, the coefficient is also negative and significant, consistent with the idea that shadow-cost changes affect returns.

Economic message: Firms with larger reductions in shadow cost have larger positive abnormal returns. This is the paper's strongest direct support for the investor-recognition mechanism.

Abnormal Returns and Changes in Shadow Cost

This table relates abnormal returns to changes in shadow cost and control variables. Two equations are specified as below:

$$AR = \alpha + \beta(dShadow) + \varepsilon$$
$$AR = \alpha + \beta_1 Log(RelSize) + \beta_2 Log(Age) + \beta_3 NYSEdum + \beta_4(dShadow) + \varepsilon$$

The dependent variable is the cumulative abnormal return from announcement to 60 days after the effective date (*CAR60*). Firm size (*RelSize*) is measured as relative to the S&P 500 index on the announcement date (firm size divided by the market cap of the S&P 500 index). The measure *Age* is defined as the number of months of listing on CRSP. The measure *NYSEdum* is set to 0 for NYSE stocks, and 1 otherwise. The measure *dShadow* is the difference between post-change shadow cost and the pre-change shadow cost as more fully described in Table V and Section IV.D. The p-values are below coefficient values in each row.

Period	Addition/ Deletion	Intercept	Log(*RelSize*)	Log(*Age*)	ExchDmy NYSE = 0	*dShadow* ($\times 10^{-9}$)	Adjusted R-square	N
197609–198909	Additions	2.564				−1.899	0.035	243
		0.011				0.002		
		−7.182	−2.393	−1.528	1.302	−1.846	0.047	243
		0.574	0.073	0.288	0.634	0.003		
	Deletions	−4.405				−14.071	−0.047	20
		0.333				0.712		
		−69.219	−5.676	2.721		−28.133	0.053	20
		0.187	0.073	0.724		0.452		
198910–200012	Additions	4.805				−0.501	0.045	186
		0.010				0.002		
		−1.981	−2.791	−2.773	2.576	−0.447	0.045	186
		0.933	0.356	0.201	0.505	0.014		
	Deletions	4.220				−24.631	0.064	54
		0.460				0.036		
		−126.003	−9.565	5.626	16.965	−27.551	0.075	54
		0.114	0.094	0.541	0.363	0.021		

6 Short summary

- The paper studies S&P 500 additions and deletions from 1962 to 2000.
- Additions generate permanent positive abnormal returns after the rise of public pre-announcements and index funds.
- Deletions generate temporary negative returns around announcement and effective dates, but these returns largely reverse.
- This asymmetry challenges symmetric demand-curve or price-pressure explanations.
- The authors propose an investor awareness explanation based on Merton's investor recognition model.
- Additions increase shareholders and institutional ownership.
- Additions reduce Merton-style shadow costs.
- Changes in shadow cost are negatively related to long-window abnormal returns.
- The findings imply that index inclusion affects investor base and required returns, not just mechanical index-fund trading demand.

7 Conclusion

7.1 Core conclusion

Chen, Noronha, and Singal show that S&P 500 index changes have asymmetric price effects. Additions create permanent price increases, but deletions do not create permanent price declines. This asymmetry is difficult to reconcile with pure price pressure or symmetric demand-shock explanations.

7.2 Economic intuition

Index addition makes a firm more visible. More investors and institutions hold or follow the stock. In a Merton-style incomplete-information setting, a larger investor base lowers the shadow cost of holding the stock and raises its price. Deletion does not erase the awareness created by past index membership, so the permanent negative effect is limited.

7.3 Takeaway

The paper reframes index effects as investor-base effects. Index additions and deletions are not mirror images because awareness is sticky: addition can create lasting recognition, while deletion need not destroy it.